

Dear Editors,

Please consider my manuscript, “Mechanistic equivalence and measurement design in models of bacterial nitrogen signaling,” for publication as a Research Article in *PLOS Computational Biology*. The work addresses a central problem in computational biology: how mechanistic models can guide experiments when distinct molecular explanations reproduce the same biological response. Its major methodological advance is an observation-conditioned guarantee that converts mechanistic non-identifiability into an experimentally actionable concentration decision, together with an end-to-end benchmark showing when complementary binding measurements can and cannot resolve the ambiguity.

Using the *Escherichia coli* GlnD–PII–GlnE–glutamine synthetase cascade, the study makes three contributions. First, it constructs biochemical mechanism families that preserve complete stationary target-state distributions while predicting different enzyme occupancies and kinetics. Second, it combines protein conservation with a concurrent PII modification-count measurement to tighten a cascade-wide bound sufficiently to justify a fivefold reduction in GlnE under the stated assay design. Third, it follows a complementary binding experiment from raw dose measurements through ligand-depletion-aware fitting and statistical discrimination, exposing how concentration bias and hidden binding-state mixtures can produce convincing but incorrect mechanistic attribution.

These results should interest the journal’s readers because they connect structural non-identifiability, biochemical network modeling and quantitative experiment design in one worked biological system. The analysis moves beyond showing that several models fit: it identifies conclusions shared across those models, the observations that separate them, and the calibration failures that limit those observations. The framework is applicable to other multisite modification cascades and signaling systems governed by bifunctional enzymes.

The manuscript presents a conditional theoretical and methodological study based on published measurements; no new experiments are reported. It clearly separates model-class guarantees from prediction error relative to biology and states the matched-assay measurements required for application. Data, executable analyses, independent numerical checks and complete manuscript sources accompany the submission.

This manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author and declare no competing interests. All published data and software reused in the study are cited and their provenance and licenses are documented.

Thank you for considering the manuscript.

Sincerely,

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